

PHY-275 Fall 2026 Pre-labs #2.

Due 9/9/26 9:00pm

You will receive 1 credit upon complete **all** the tasks.

Task #1: Compute eigenvalues from the state-space matrix form.

$$\ddot{x} + \dot{x} + 2x = 0 \quad \text{I. C.: } x_0 = 2, \dot{x}_0 = -3$$

$$\ddot{x} + \dot{x} - 2x = 0 \quad \text{I. C.: } x_0 = 3, \dot{x}_0 = 0$$

Pre-lab for next week: to use state-space matrix (i.e. coupled 1st order matrix of equations) to write these two equations and compute the eigenvalues.

To refresh your memory, this is how you can compute eigenvalues from matrix form (https://en.wikipedia.org/wiki/Eigenvalues_and_eigenvectors):

As a brief example, which is described in more detail in the examples section later, consider the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Taking the determinant of $(A - \lambda I)$, the characteristic polynomial of A is

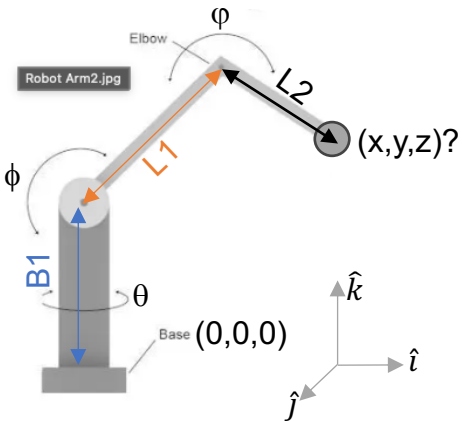
$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = 3 - 4\lambda + \lambda^2.$$

Setting the characteristic polynomial equal to zero, it has roots at $\lambda = 1$ and $\lambda = 3$, which are the two eigenvalues of A .

As a proof-of-efforts, write down each viable steps, and upload as a single file into Moodle.

Task #2: With the help of Sequential Euler rotations in 3D (slide #7), derive the DH parameters for 3D forward kinematics, two arms and two joints.

No need to compute the final matrix, just matrix for each rotation (joint) and translation (arm).



As a proof-of-efforts, write down each viable steps, and upload as a single file into Moodle.